

Statistical Word Segmentation: Emergent Structure in a Next-Character RNN

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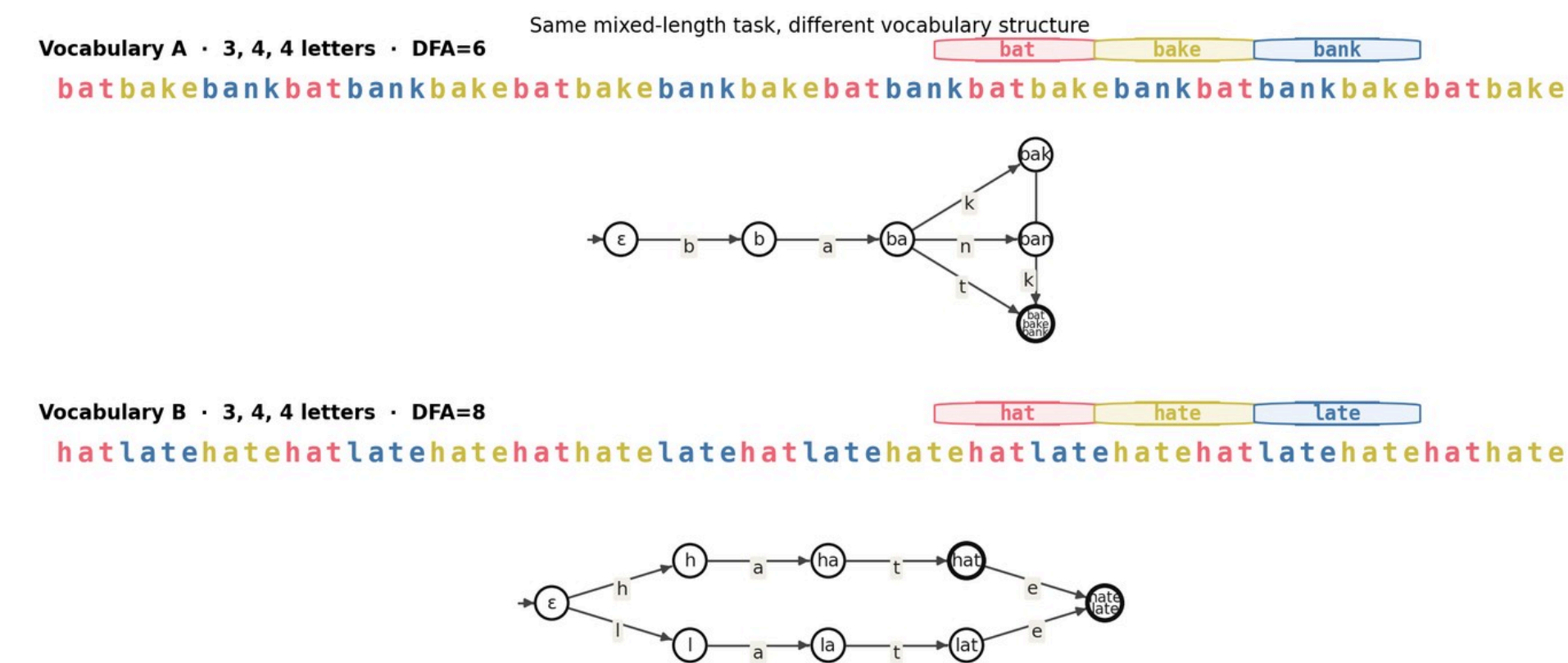
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Abstract

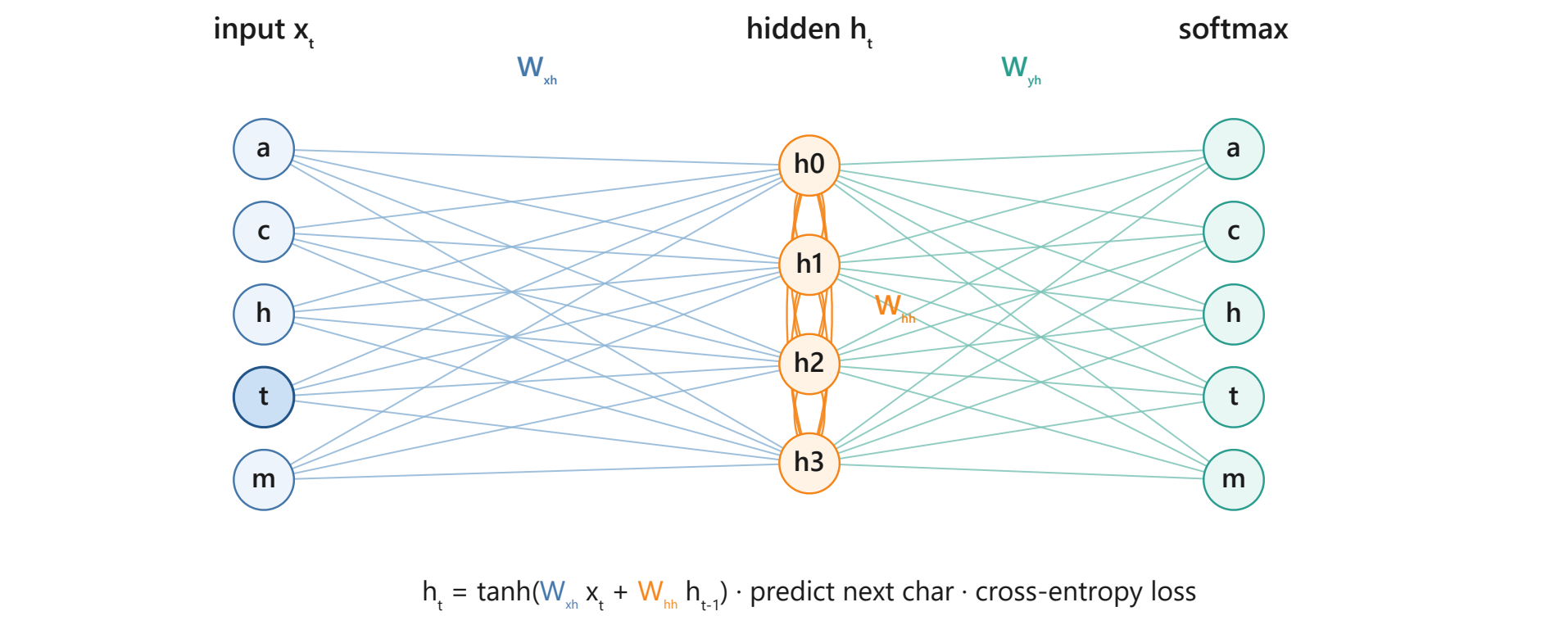
Eight-month-old infants can segment continuous speech by tracking transitional probabilities (Saffran, Aslin & Newport, 1996). We ask whether a vanilla Elman RNN trained only on next-character prediction develops internal representations aligned with word structure. After learning, the network generates legal vocabulary items, and its hidden states become a continuous embedding of the vocabulary's minimal DFA. On an eight-word -ate/-at demo, DFA state explains $\eta^2 \approx 0.88$ of condensed hidden variance; word identity is much weaker ($\eta^2 \approx 0.06$). Fifty mixed-length English vocab runs ($H=100$) show that dimensionality, training cost, and readouts track minimized DFA size. Weight matrices become letter-columnar in W_{in} and locally clumped in W_{hh} , most clearly for small automata.

Fig 1 · Mixed vocabularies & minimal DFAs



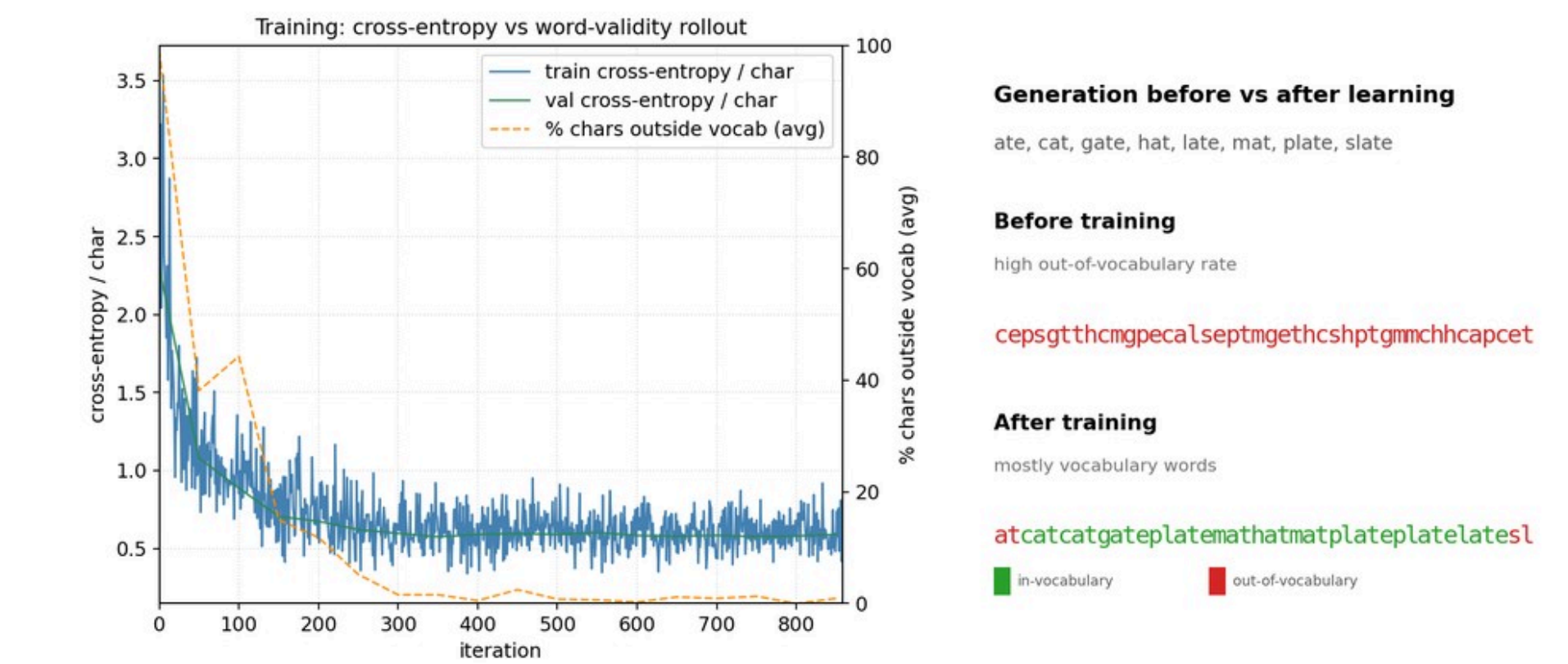
Two ≤ 3 -word conditions. Word chips, one unsegmented stream, then the prefix-labeled minimal DFA (6 vs 8 states), laid out left to right.

Fig 2 · Training the RNN



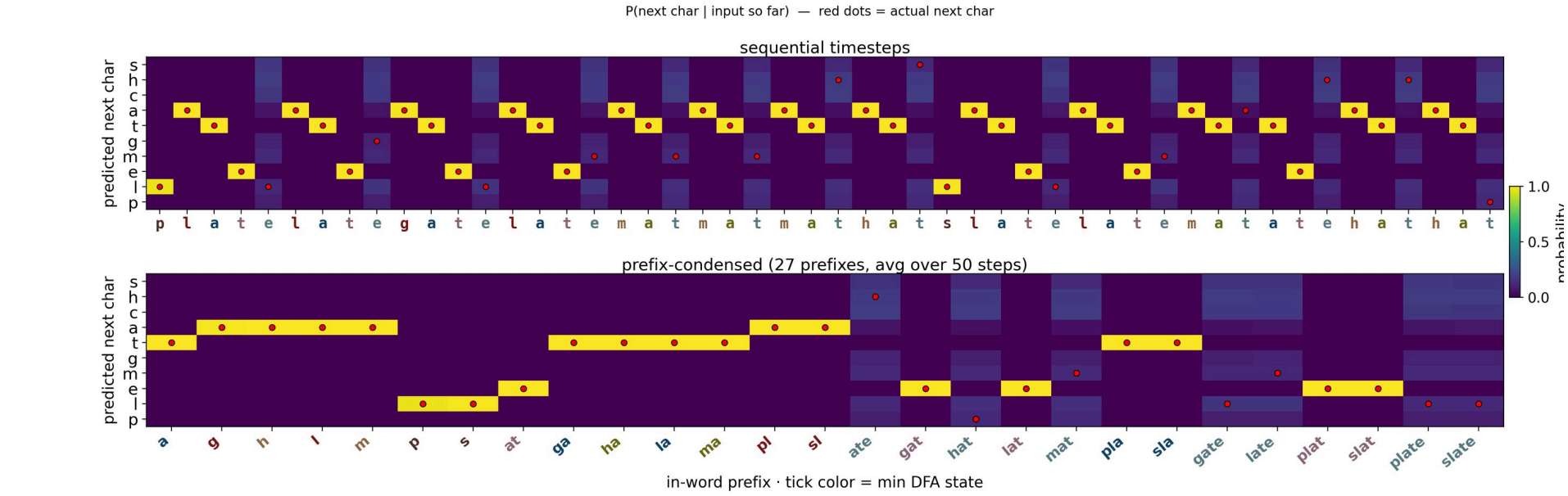
Character-level Elman RNN. Input x_t predicts the next character via $h_t = \tanh(W_{in}x_t + W_{hh}h_{t-1})$; cross-entropy loss.

Fig 3 · Learning & generation



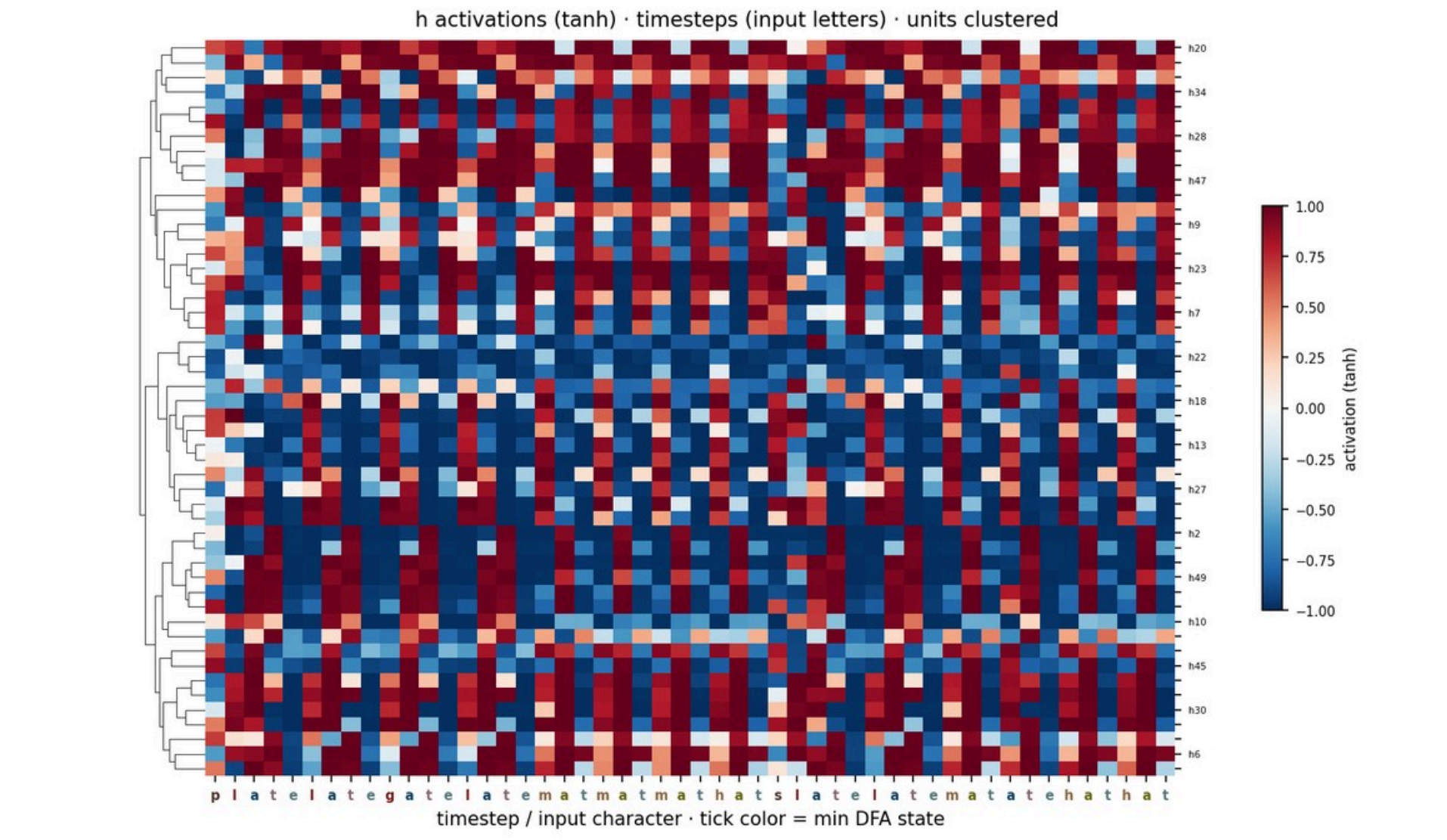
Loss and OOV collapse together; rollouts become legal vocabulary words.

Fig 4 · Next-character probabilities



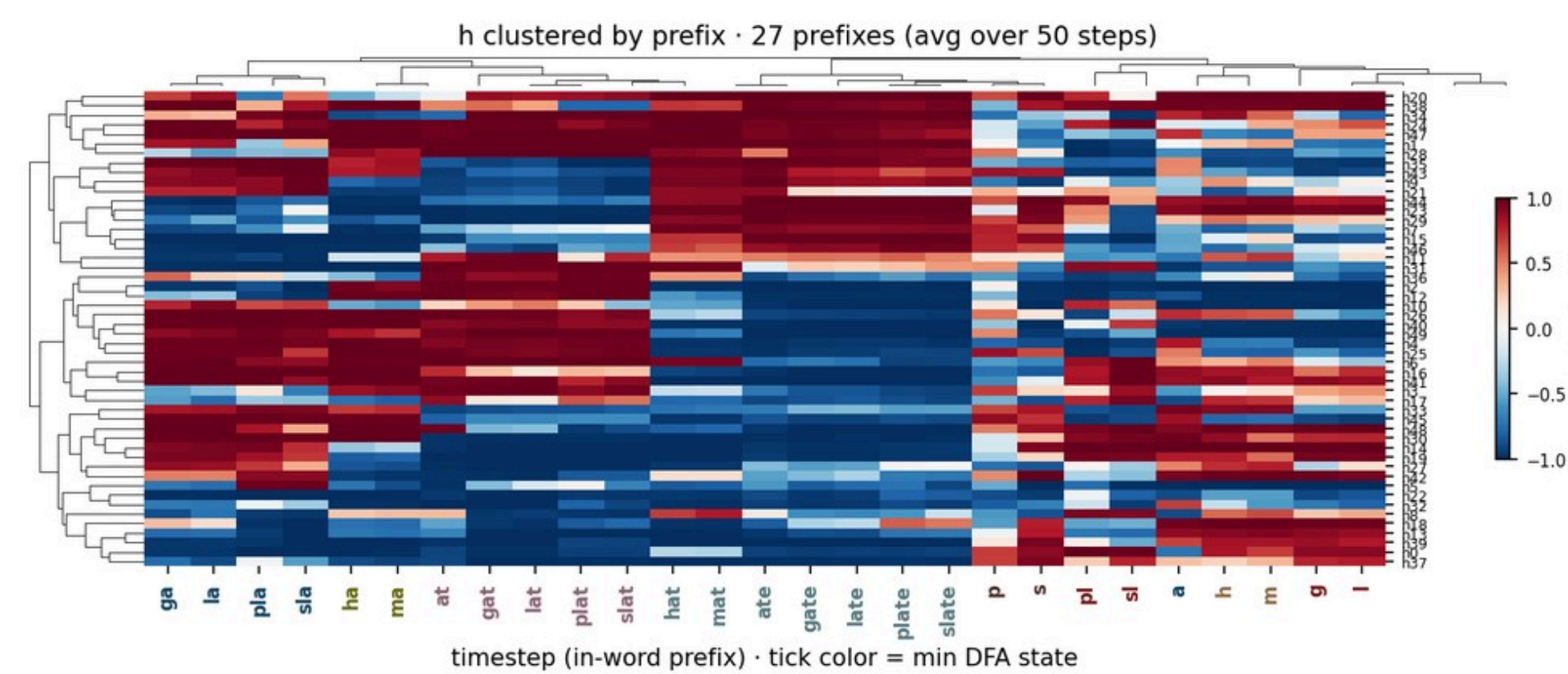
Softmax $P(\text{next char} | \text{input so far})$. Red dots = actual next character.

Fig 5 · Activation heatmap



Hidden units over timesteps (input letters; units clustered).

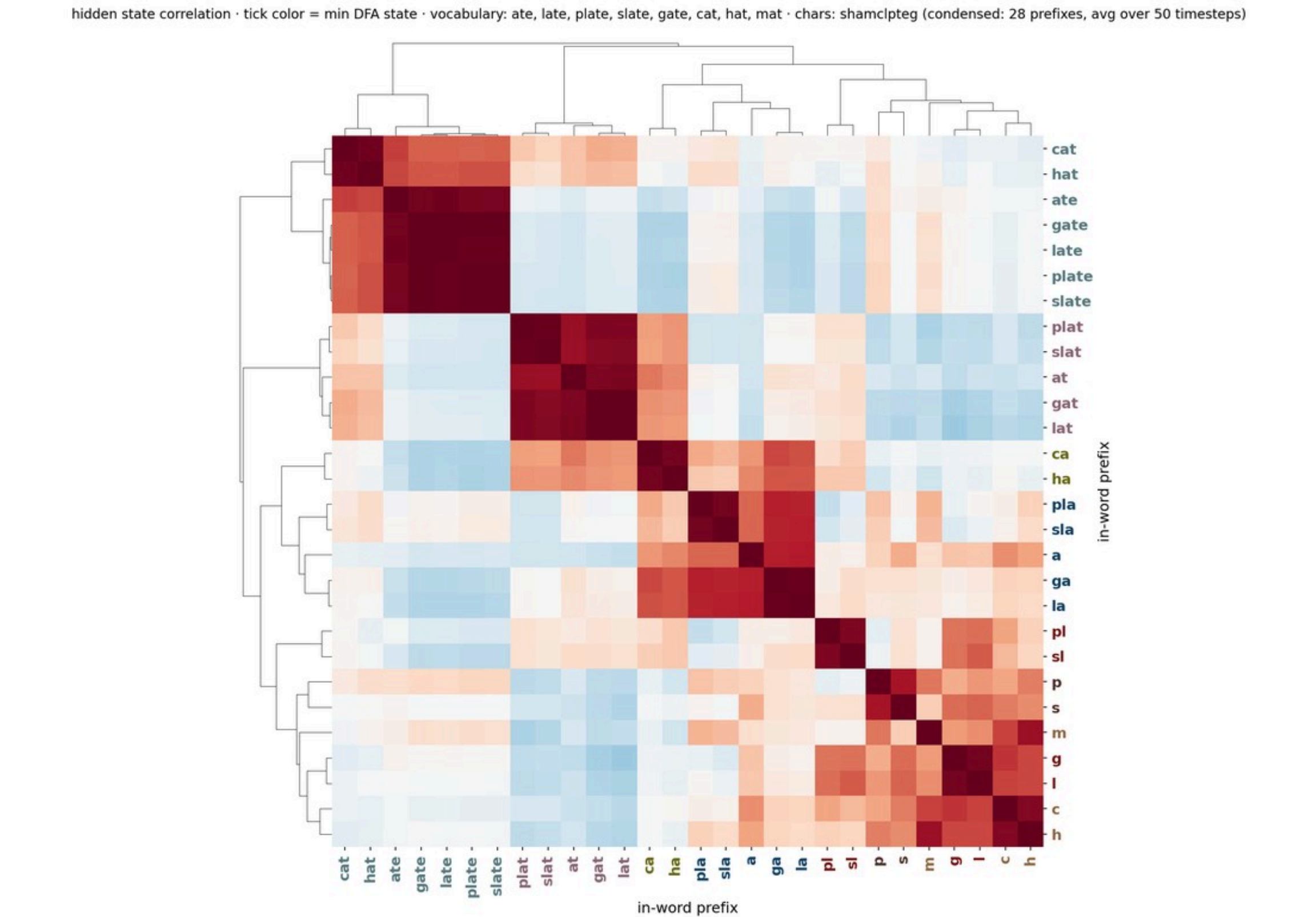
Fig 6 · Activations by prefix



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Key references: Saffran, Aslin & Newport (1996); Elman (1990); Aslin, Saffran & Newport (1998); Goldwater et al. (2009); Perruchet & Vinter (1998).

Fig 7 · Hidden-state correlation



Prefix $\times$ prefix correlation. Blocks merge prefixes that share a future.

Fig 8 · DFA geometry & separation

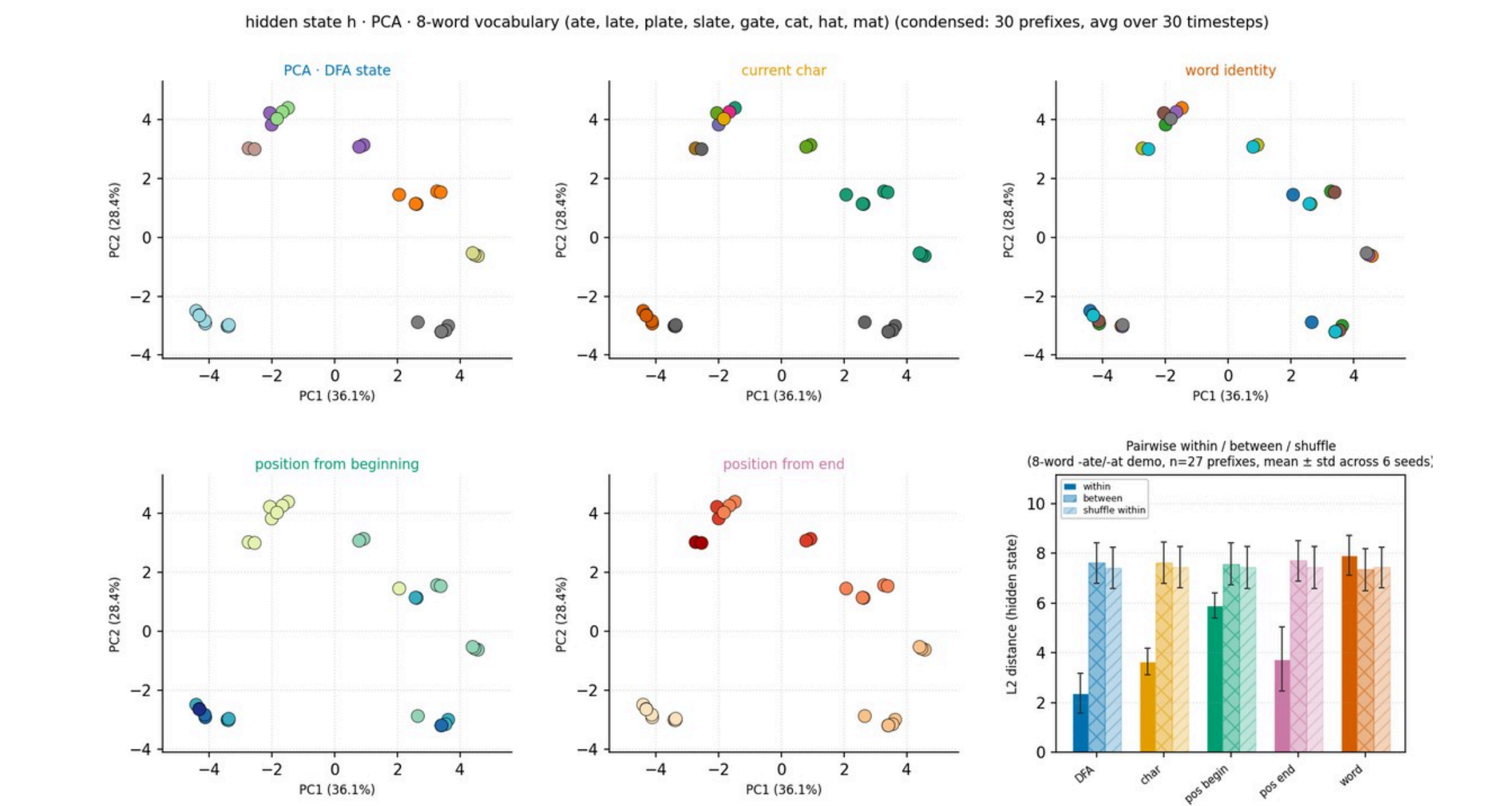


Fig 9 · Example selective units

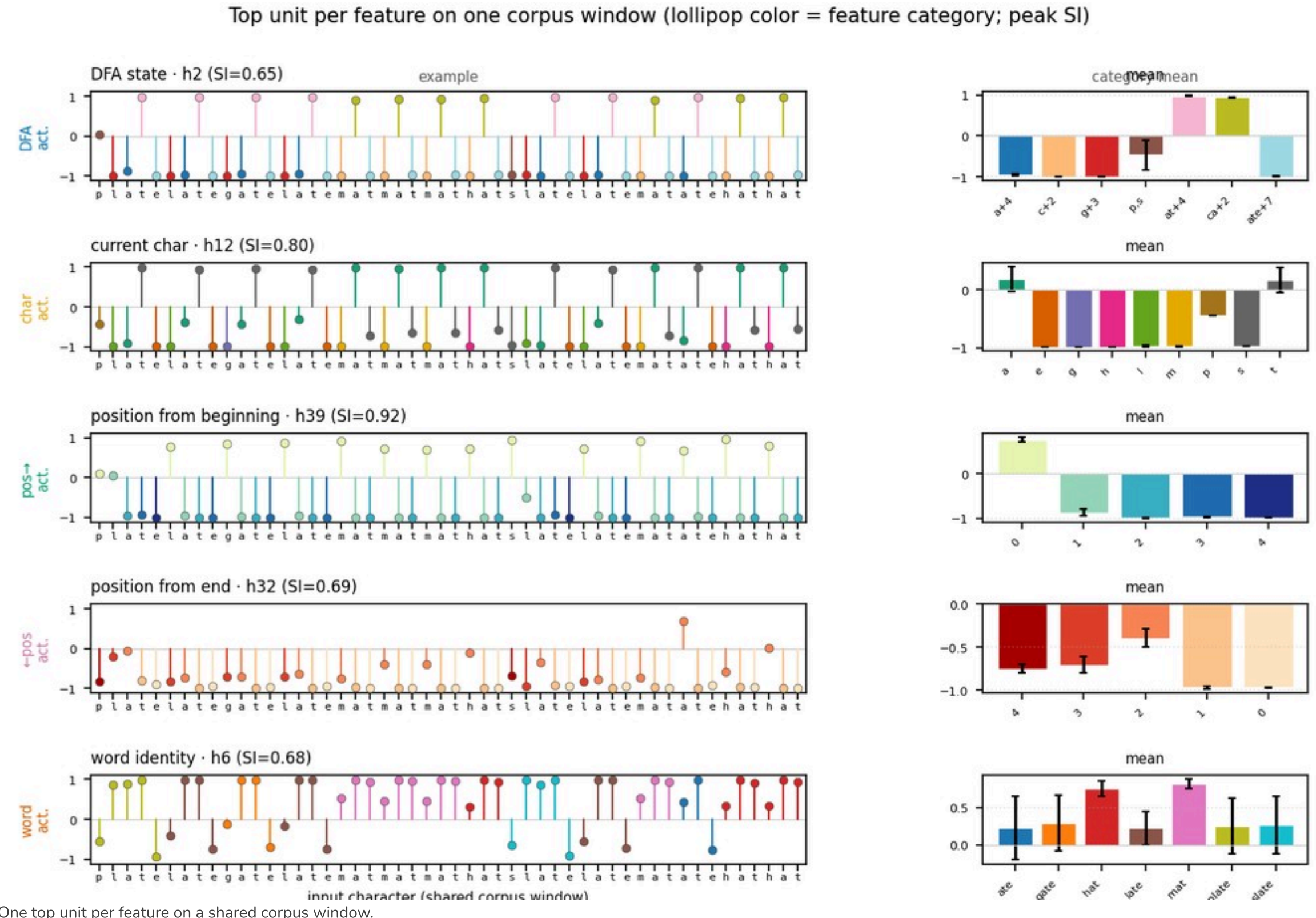


Fig 10 · Linear decoding

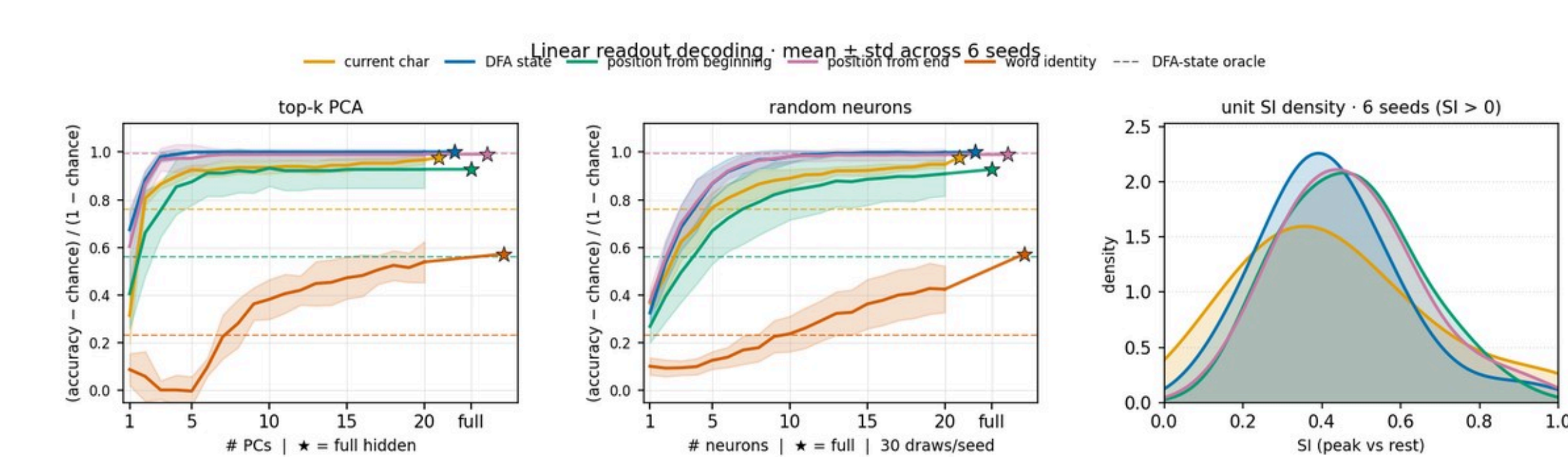
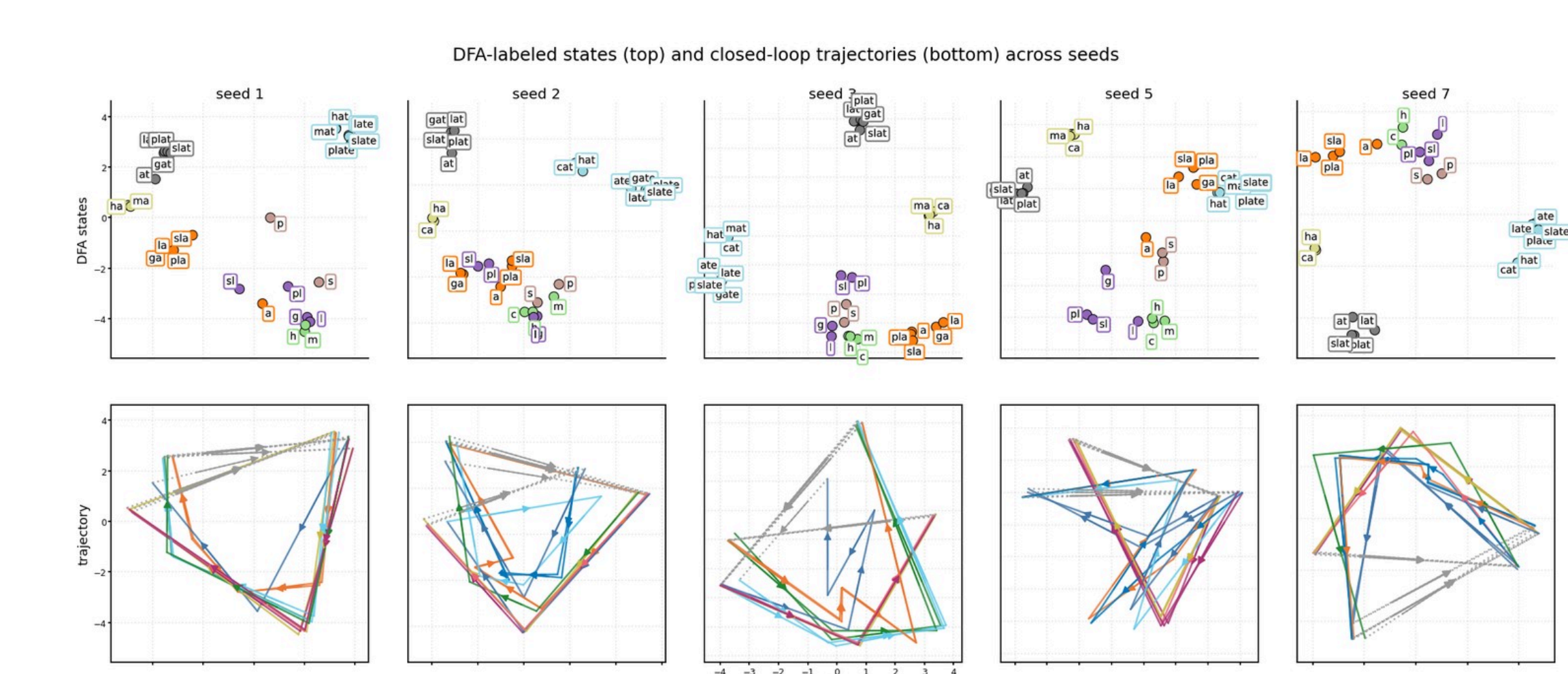


Fig 11 · Word trajectories



DFA-labeled PCA (top) and closed-loop trajectories (bottom) across five seeds.

Fig 12 · Mixed vocabs track DFA size

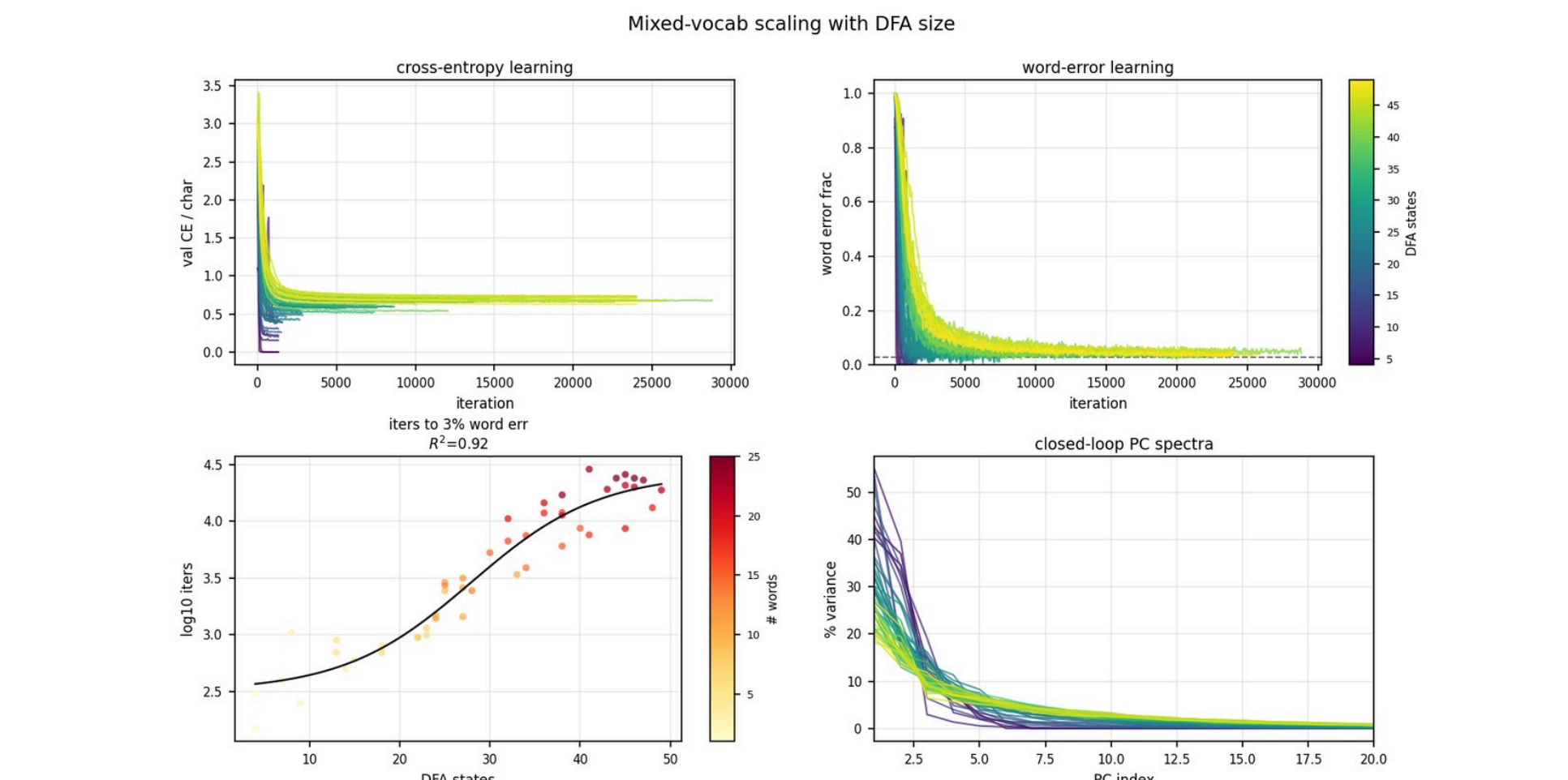


Fig 13 · Activations vs DFA size

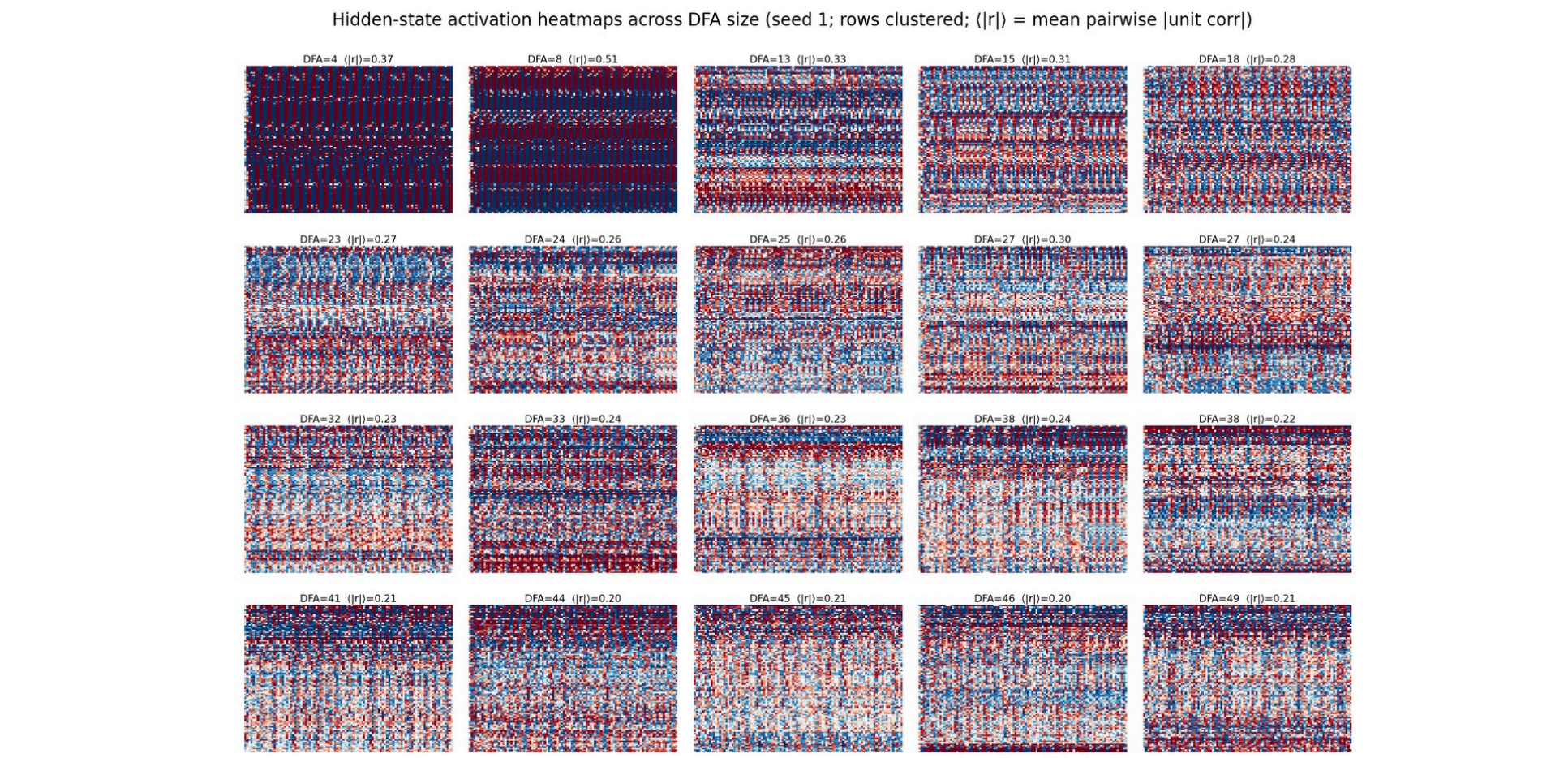


Fig 15 · Readouts vs DFA size

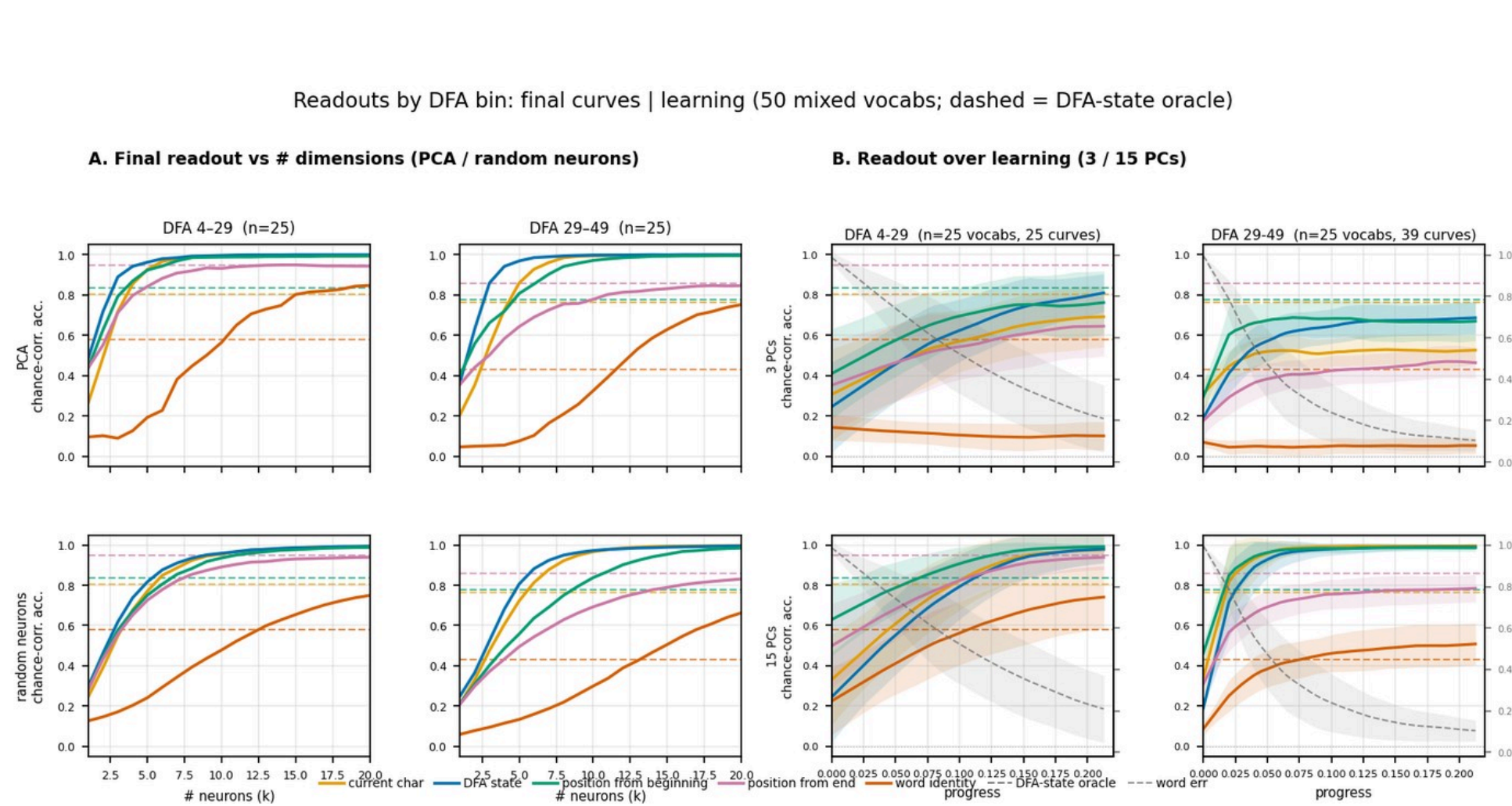


Fig 16 · Geometry & unit selectivity

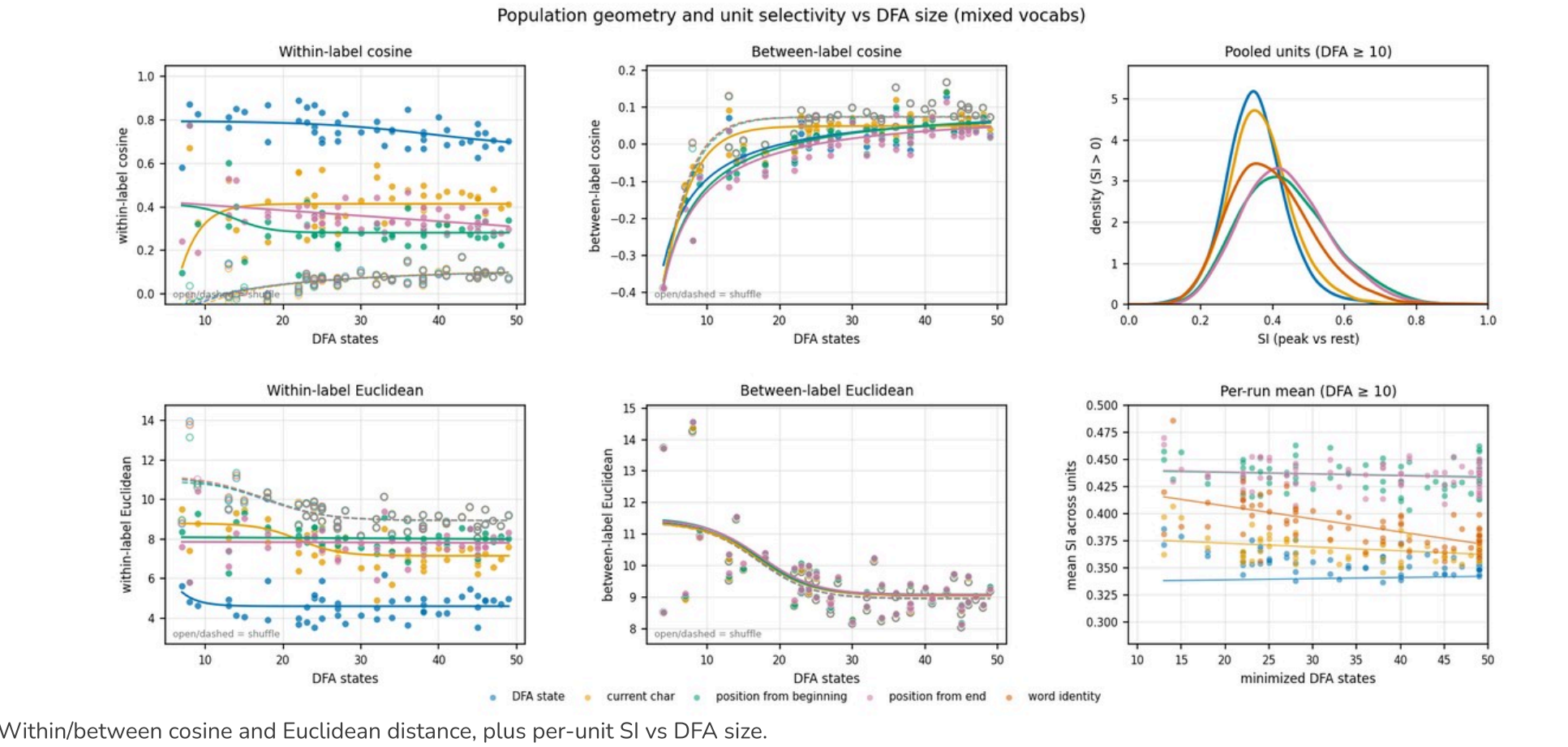


Fig 18 · Weight matrices vs DFA

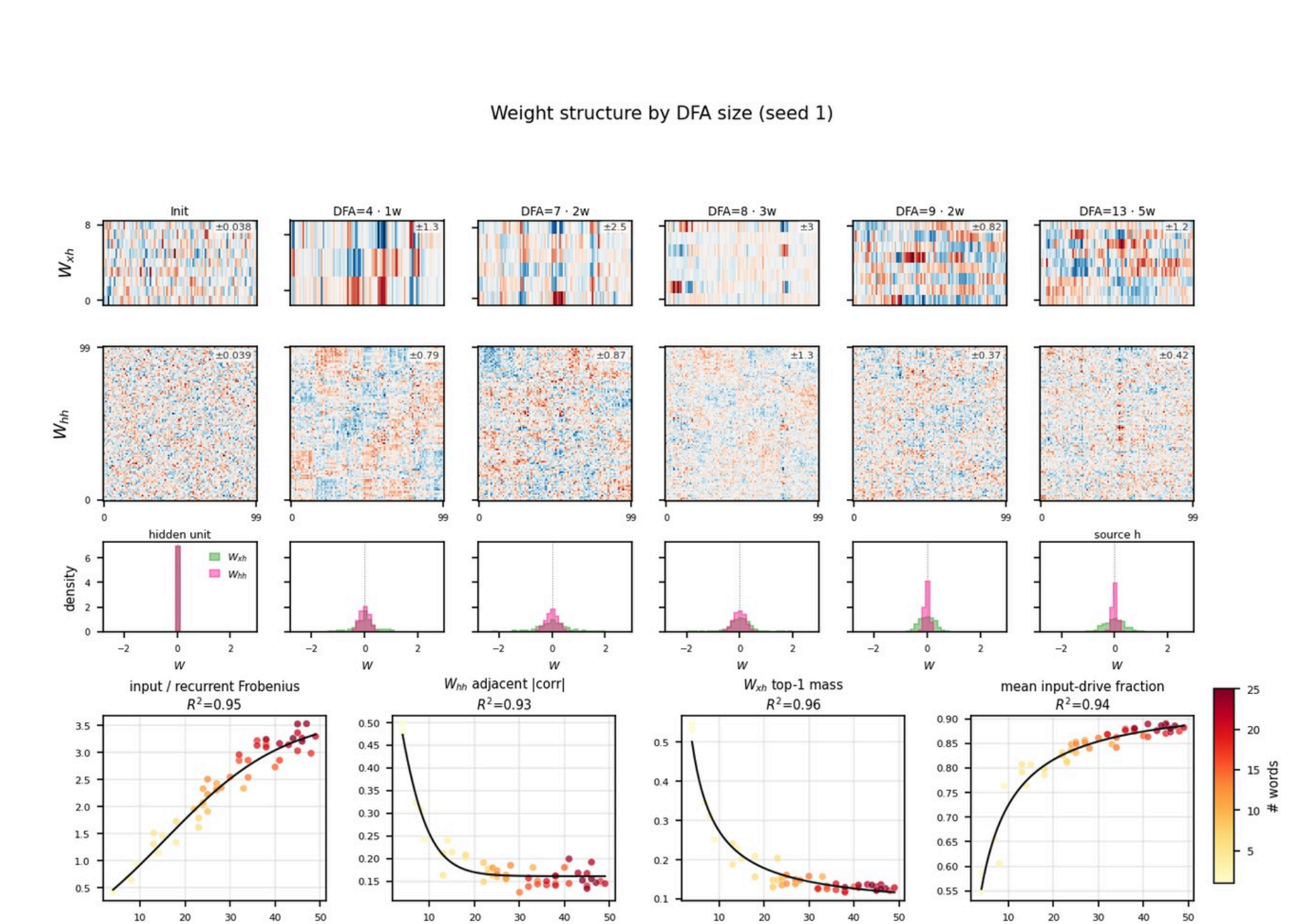


Fig 19 · Weight graph motifs vs DFA

